

Scientific Computation

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Lecture 4 - Cancellations

IEEE 754 Visualizations

Really useful resource:

<https://bartaz.github.io/ieee754-visualization/>

IEEE 754 Overview

We describe a FP system with (β, t, L, U)

For example, $(\beta = 10, t = 4, L = -2, U = 2)$ gives us
numbers like 1.207×10^{-1}

But we've glossed over the particulars of a few important
concepts: normalized/denormalized numbers, sticky bit, and the
biased exponent

Let's **very carefully** look at the details of how IEEE 754 numbers
are represented

IEEE 754 Overview

Interpreting the value of a floating point number $x = (\text{sign}, e, d)$
depends on the value of the **biased exponent** bits e .

Case 1 - Normalized Numbers

If $e \neq 0$, we have a **normalized** floating point number:

$$x = (-1)^{\text{sign}} \times \beta^{e-\text{offset}} \times (1 + \sum_{i=1}^{t-1} d_i \beta^{-i})$$

$$= (-1)^{\text{sign}} \times \beta^{e-\text{offset}} \times (1.d_1 d_2 \dots d_{t-1})$$

Bias = 1023 for doubles

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IEEE 754 Overview

Interpreting the value of a floating point number $x = (\text{sign}, e, d)$ depends on the value of the **biased exponent** bits e .

Case 2 - Denormalized Numbers

If $e = 0$, we have a **denormalized** floating point number:

$$x = (-1)^{\text{sign}} \times \beta^{e-\text{offset}+1} \times (0 + \sum_{i=1}^{t-1} d_i \beta^{-i})$$

Smallest
↑

$$= (-1)^{\text{sign}} \times \beta^{e-\text{offset}+1} \times (0.d_1 d_2 \dots d_{t-1})$$

The sticky bit (d_0) is “free” because it is always determined by the value of the exponent e !

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IEEE 754 Overview

Interpreting the value of a floating point number $x = (\text{sign}, e, d)$ depends on the value of the **exponent** bits e .

Case 3 - Exceptional Values

If the bits of e are all ones and the mantissa (d) is all zeros, then we have $\pm\text{Inf}$

If the bits of e are all ones and there is at least one nonzero value in the bits of the mantissa, then we have NaN

Finally, if all bits (except sign) are zero, we have ± 0 (some conventions consider zero a denormalized number)

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Lecture Outline

- Cancellations
- History

Cancellations

Consider $x - y$, where $x \neq y$ are real numbers.

Assume there is no roundoff in the subtraction, i.e.,

$$fl(x - y) = fl(x) - fl(y) \text{ (best case scenario for subtraction)}$$

We know $fl(x) = x(1 + \epsilon_1)$ and $fl(y) = y(1 + \epsilon_2)$.

$$\begin{aligned} fl(x - y) &= fl(x) - fl(y) & |\epsilon_1| \leq u, \quad |\epsilon_2| \leq u \\ &= x(1 + \epsilon_1) - y(1 + \epsilon_2) \\ &= x - y + x\epsilon_1 - y\epsilon_2 \\ &= (x - y) \left[1 + \frac{x\epsilon_1 - y\epsilon_2}{x - y} \right] \\ &= (x - y)(1 + \Sigma), \text{ where } \Sigma = \frac{x\epsilon_1 - y\epsilon_2}{x - y} \end{aligned}$$

Σ is the relative error in $fl(x - y)$

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Addition vs. Subtraction

Example 1: Consider a decimal FP system with $t = 5$ digits.

Let $x = 9.23450001$ and $y = 9.23455001$.

Assuming rounding to the nearest FP, what is the relative error in:

(a) $fl(x + y)$, and (b) $fl(x - y)$?

$$u = \frac{\epsilon_{\max}}{2} = \frac{1.0001 - 1}{2} = 5 \times 10^{-5}$$

$$\begin{aligned} \text{(a) } fl(x + y) &= fl[fl(x) + fl(y)] = fl(9.2345 + 9.2346) \\ &= fl(1.84691 \times 10^1) \\ &= 1.8469 \times 10^1 \end{aligned}$$

$$\begin{aligned} |\epsilon| &= \left| \frac{fl(x + y) - (x + y)}{x + y} \right| = \left| \frac{1.8469 \times 10^1 - 1.84690002 \times 10^1}{1.846905002 \times 10^1} \right| \\ &= 2.7 \times 10^{-6} < 5 \times 10^{-5} \triangleq u \end{aligned}$$

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Addition vs. Subtraction

Example 1 continued: Consider a decimal FP system with $t = 5$ digits.

Let $x = 9.23450001$ and $y = 9.23455001$.

$$\begin{aligned}
 (b) \text{ fl}(x - y) &= \text{fl}[\text{fl}(x) - \text{fl}(y)] \\
 &= \text{fl}[9.2345 - 9.2346] \\
 &= \text{fl}[-1.0000 \times 10^{-4}] \\
 &= -1.0000 \times 10^{-4} \quad \text{No sub. rounding error}
 \end{aligned}$$

$$\begin{aligned}
 \left| \frac{\text{fl}(x-y) - (x-y)}{x-y} \right| &= \left| \frac{-1 \times 10^{-4} - (-5.0000 \times 10^{-5})}{-5 \times 10^{-5}} \right| \\
 &= \left| \frac{-10 - (-5)}{-5} \right| \\
 &= 1 > 5 \times 10^{-5} \triangleq u
 \end{aligned}$$

Catastrophic cancellation 2 numbers are closed in "-"

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Cancellations

Cancellations occur when subtracting *nearby* numbers that contain roundoff.

Example 2: Assume a decimal FP system with $t = 5$ digits and rounding to nearest. Let $x = \underline{1.234567}$ and $y = \underline{1.234512}$ and compute $x - y$ in this FP system.

$$\text{fl}(x) = 1.2346, \text{ fl}(y) = 1.2345$$

$$\text{fl}(x) - \text{fl}(y) = 0.0001 \rightarrow \text{this error is due to rounding, the subtraction was exact}$$

$$x - y = 5.5 \times 10^{-5}$$

Cancellations

Example 2 continued

The absolute error in $f(x) - f(y)$ is small,

$$[f(x) - f(y)] - (x - y) = 4.5 \times 10^{-5}$$

The relative error is:

$$\frac{\text{abs error}}{x - y} = \frac{4.5 \times 10^{-5}}{5.5 \times 10^{-5}} = 0.82 = 82\%$$

Cancellations

$t=5$

Example 3: Now let $x = 5.384576$ and $y = 4.894080$

$f(x) = 5.3846$, $f(y) = 4.8941$ the digits 6 and 1 contain round off error

$$f(x) - f(y) = 0.4905$$

No round off error introduced by subtraction

$$= \underbrace{4.905}_{\text{correct in first 3 digits}} \times 10^{-1}$$

Error from $f(x)$ and $f(y)$

Cancellations

Example 3 continued

True result is $x - y = 5.384376 - 4.894080$
 $= 0.490296$

Abs. error is $fl(x) - fl(y) - (x - y) = 4.0000 \times 10^{-6}$

Relative error is $\frac{\text{abs error}}{x - y} \approx 8.16 \times 10^{-6}$
 $u = 5 \times 10^{-5}$

Avoiding Cancellations

Example 4: Consider the equivalent expressions $x^2 - y^2$ and $(x - y)(x + y)$. Suppose $|x| \approx |y|$. Which one is better to evaluate? Assume $x, y > 0$; the case $x, y < 0$ is similar.

- $x - y$ is at risk of catastrophic cancellation
- $x + y$ is no issue ($|z| \approx u$)
- x^2 and y^2 are FP operations so we still be introduced

- a very small operation, so the relative roundoff error in $\text{fl}(x^2)$ and $\text{fl}(y^2)$

Because of the additional risk of roundoff in the $\sqrt{}$ operation, we conjecture that $x^2 - y^2$ might be worse than $(x+y)(x-y)$

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Avoiding Cancellations

Example 4 continued: The “BigFloat” type in Julia uses arbitrary precision floating points to get more accurate representations - good way to check our analytical predictions!

```
using Random
```

```
x = 10000*rand()
y = x * (1 + 1e-10);
eval1 = (x - y)*(x + y)
eval2 = x^2 - y^2
xv = BigFloat(x)
yv = BigFloat(y)
acc = (xv - yv)*(xv + yv)
```

```
println("Rel. error in (x-y)*(x+y) = $((acc-eval1)/acc)")
println("Rel. error in x^2 - y^2 = $((acc-eval2)/acc)")
```

```
✓ 0.0s
```

```
Rel. error in (x-y)*(x+y) = 1.645417679222518318011807033397011039229323014017176671822896658779294315625732e-16)
Rel. error in x^2 - y^2 = -4.0927791953089720186172752846548556257207027887017475034487090504557966333784e-07)
```

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Summary

Three ingredients for cancellation to occur:

1. Subtraction

2. Close values

3. Round off error is present in the values being subtracted

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Lecture Outline

- Cancellations
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Ariane 5

Recall the disaster video from the first lecture – the June 4th, 1996 launch

Recall the disaster video from the first lecture - the June 4th, 1990 launch of the Ariane 5 rocket.

A 64-bit floating point number was converted to a 12-bit integer!
Overflow checking was limited for performance reasons, and assumptions from the Ariane 4's flight were incorrectly made



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Vancouver Stock Exchange

- In 1982, the Vancouver Stock Exchange started an electronic stock index set initially to 1000 points
- The index was updated after each transaction
- In 22 months the index **fell to 520**
- It was not supposed to fall in a bull market!
- Investigation showed each intermediate result was rounded to **2 decimals** by chopping, e.g., 568.958 rounds to 568.95
- Once corrected, the index was at **1098.892**

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在这章我们一般忽略了subtraction roundoff error, 这是因为我们减法得到的结果可以被
当差的系统表示, 当出现那些不能转成二进制的小数时, 它们会经过50, 100, 200, 400, 800, 1600, 3200, 6400, 12800, 25600, 51200, 102400, 204800, 409600, 819200, 1638400, 3276800, 6553600, 13107200, 26214400, 52428800, 104857600, 209715200, 419430400, 838860800, 1677721600, 3355443200, 6710886400, 13421772800, 26843545600, 53687091200, 107374182400, 214748364800, 429496729600, 858993459200, 1717986918400, 3435973836800, 6871947673600, 13743895347200, 27487790694400, 54975581388800, 109951162777600, 219902325555200, 439804651110400, 879609302220800, 1759218604441600, 3518437208883200, 7036874417766400, 14073748835532800, 28147497671065600, 56294995342131200, 112589990684262400, 225179981368524800, 450359962737049600, 900719925474099200, 1801439850948198400, 3602879701896396800, 7205759403792793600, 14411518807585587200, 28823037615171174400, 57646075230342348800, 115292150460684697600, 230584300921369395200, 461168601842738790400, 922337203685477580800, 1844674407370955161600, 3689348814741910323200, 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592855496895058920568683443244